

Hagia Sophia

Rebuilt on the orders of Byzantine emperor Justinian from 537 CE after the original burnt to the ground, Constantinople's Hagia Sophia, or Church of the Holy Wisdom, was one of the most lavish and expensive buildings of all time. Designed by scientists Anthemius of Tralles and Isidore of Miletus, it took more than 10,000 workmen five years to build.

Its greatest triumph is a 32.5-metre (107-ft) dome, raised over a square central space using tapering triangular elements known as pendentives. This technique allowed the builders to dispense with supporting walls, instead using half-domes to the east and west to hold up the main dome (heavy reinforcing buttresses were added later). The dome was overlaid in gold to evoke Heaven, with the square space below laid in coloured marble symbolizing Earth. With the decline of the Byzantine Empire, the building was ransacked several times, and when the Ottoman Turks stormed Constantinople in 1453, Sultan Mehmed II converted it into a mosque, with minarets added over the next century.

